DNN PYQ Volume 1 - Solved Questions

Q1 CNN Output Shape

Input 8x8, kernel 3x3, stride=1, padding=0. Formula: (N-F+2P)/S+1=(8-3+0)/1+1=6. Output=6x6. With 5 filters => 6x6x5. Pooling 4x4 stride2 => ((6-4)/2)+1=2. Final output=2x2x5.

Q2 Gradient Descent

E=3w1^2+4w2^2+2w1w2. dE/dw1=6w1+2w2. dE/dw2=8w2+2w1. At (0.5,0.5): gradients=(4,5). New weights with lr=0.1: w1=0.5-0.4=0.1, w2=0.5-0.5=0.

Transfer Learning

Small dataset training from scratch overfits. Transfer learning uses pretrained features, reduces data need and improves generalization.

Perceptron Template

1. Compute z=w.x+b 2. Apply step activation 3. Compare with label 4. Update w_new=w+eta(y-yhat)x.

Logistic Regression Template

1. z=wTx+b 2. yhat=sigmoid(z) 3. error=yhat-y 4. gradient=(yhat-y)x 5. update w=w-lr*gradient.

Softmax Template

Compute exp(zi), divide by sum, choose largest probability, CCE=-log(true class prob).

Forward Propagation

Z1=W1X+b1; A1=g(Z1); Z2=W2A1+b2; A2=sigmoid(Z2).

Backpropagation

dZ2=A2-Y; dW2=dZ2A1^T; dZ1=(W2^T dZ2)*g'(Z1); dW1=dZ1X^T.

Weak Areas

Jacobian: matrix of partial derivatives. Chain rule: multiply gradients layer by layer.

Practice Problem 1

Solve fully without notes and verify formulas.

Practice Problem 2

Solve fully without notes and verify formulas.

Practice Problem 3

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Practice Problem 4

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Practice Problem 5

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Practice Problem 6

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Practice Problem 7

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Practice Problem 8

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Practice Problem 9

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Practice Problem 10

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Practice Problem 11

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Practice Problem 12

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Practice Problem 13

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Practice Problem 14

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Practice Problem 15

Solve fully without notes and verify formulas.